

TwinSanity Recon

AI-Powered Web Reconnaissance &
Vulnerability Assessment Platform

v2.0.0

Python 3.10+

MIT License

github.com/MustafaCyb/TwinSanity-Recon



What is TwinSanity Recon?

A comprehensive, AI-powered platform for security professionals and bug bounty hunters

TwinSanity Recon is a web-based reconnaissance and vulnerability assessment platform that combines traditional security scanning with modern AI-powered analysis.

It aggregates multi-source data into a single, unified workflow — from subdomain discovery to CVE detection to AI-generated security reports.

Designed For

Bug Bounty Hunters
Penetration Testers
Enterprise Security Teams
Organizations with large subdomain estates

10+

Subdomain Sources

5+

AI LLM Providers

3

User Roles

MIT

Open Source License

Python 3.10+ | Flask Web Dashboard
Multi-LLM AI | Ollama Local Support

Reconnaissance Capabilities

Passive collection from 10+ sources with comprehensive scanning modules

Subdomain Enumeration

Passive collection from crt.sh, SecurityTrails, VirusTotal, Chaos, HackerTarget, AlienVault OTX and more

DNS Resolution

Async DNS resolution with multi-record support (A, AAAA, CNAME, MX, NS, TXT)

HTTP Probing

Live host detection with status codes, titles, technology fingerprinting via headers

Port Scanning

Configurable port scanning with service detection and banner grabbing

CVE Detection

Automated vulnerability scanning with CVE lookup via NVD API and NIST references

URL & API Harvesting

Endpoint discovery from Wayback Machine, CommonCrawl, and web scraping

AI-Powered Analysis

Multi-LLM support with selectable providers and local Ollama integration

Multi-LLM Providers

Supports 5+ LLM providers — select and block providers via the web dashboard. Configure API keys in config.yaml for maximum flexibility.

Local AI with Ollama

Run LLMs locally with Ollama — the dashboard auto-detects all installed local models. Complete privacy with zero cloud dependency.

AI Security Assistant

Built-in AI chat assistant to discuss findings, analyze vulnerabilities, and get remediation advice in real time.

What AI Analyzes

- Scan results interpretation
- Vulnerability risk assessment
- Subdomain CVE discovery
- Attack surface mapping
- Remediation recommendations
- Security report generation
- NIST/NVD CVE references

Web Dashboard & Features

Modern Flask-based web interface with real-time updates and integrated tools

● WebSocket Live Updates

Real-time scan progress with live subdomain, DNS, and vuln results

● Integrated Shodan Engine

Built-in Shodan search for exposed services and device intelligence

● CVE Explorer

Browse and search CVEs with NIST NVD references and severity ratings

● Admin Panel

User management with 3 roles, scan history, and system configuration

● AI Chat Interface

Interactive AI assistant for discussing and analyzing scan findings

● Light & Dark Mode

Full theme support — dark mode by default with light mode option

Scan Results & Reporting



Recon Phase

- Subdomain enum (10+ sources)
- DNS multi-record types
- HTTP probing + tech fingerprint
- Port scanning + URL harvesting

CVE Lookup Phase

- CVE detail + NIST NVD links
- CVSS severity scoring per finding
- CVE references linked to NVD database
- Automated CVE matching to findings

AI Analysis Phase

- GPT / Claude / Gemini / etc.
- Ollama local LLM (no cloud)
- Risk scoring per finding
- AI chat for findings review

Scan Results Viewer

Organized findings with filtering, search, and AI-driven insights

- Filterable & searchable findings dashboard
- Subdomain CVE discovery with severity tags
- Public scan sharing for team collaboration
- TwinSanity Island - interactive result map

Report Generation

Auto-generated reports with executive & technical sections

- Executive summary + technical detail sections
- Vulnerability severity breakdown (CVSS)
- CVE references directly linked to NIST/NVD
- Actionable remediation recommendations

Architecture & Tech Stack

Modular Python architecture with configurable modules and API integrations

Tech Stack

Language Python 3.10+



Backend Flask + WebSocket



Frontend HTML/CSS/JS Dashboard

AI Engine Multi-LLM + Ollama



CVE Data NVD API Integration

Scanning Async Multi-Module

License MIT Open Source



config.yaml Setup

Simple YAML configuration for API keys, module selection, and LLM provider settings. Add or block modules and providers directly from config.

Modular Design

Each recon module can be enabled or disabled independently. Custom module names displayed in the UI. Extensible architecture for adding new scanning capabilities.

Quick Start

Get up and running in minutes with Python and a virtual environment

```
● ● ● terminal – install & run
```

Clone the repository

```
git clone github.com/MustafaCyb/TwinSanity-Recon.git  
cd TwinSanity-Recon
```

Create virtual environment

```
python -m venv venv  
source venv/bin/activate
```

Install dependencies

```
pip install -r requirements.txt
```

Run the dashboard

```
python run_dashboard.py
```

1

Configure

Edit config.yaml with your API keys for SecurityTrails, VirusTotal, Shodan, and NVD

2

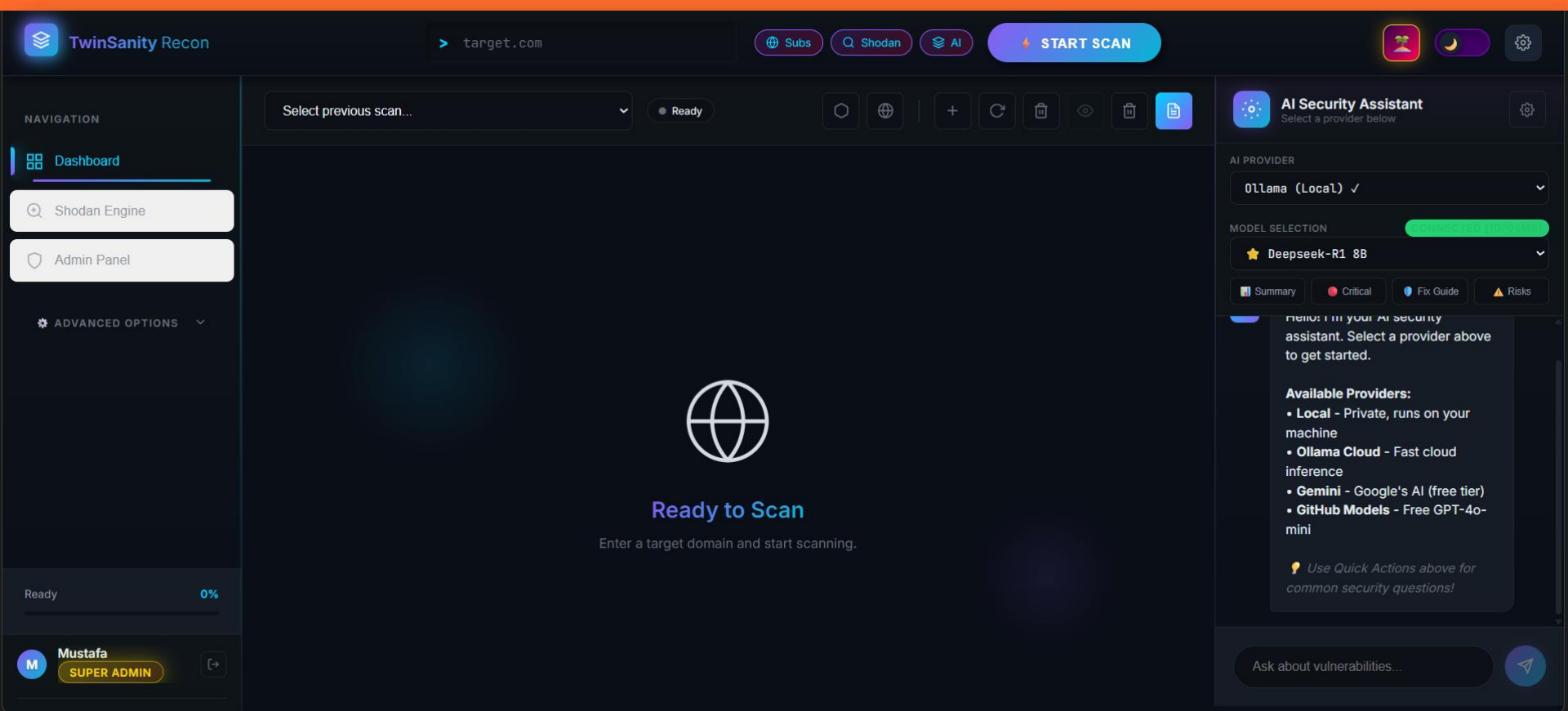
Launch

Run python run_dashboard.py to start the Flask web server on localhost

3

Scan

Enter a target domain and start your reconnaissance via the web UI



MAIN WEB UI

TwinSanity Recon

> target.com

Subs

Shodan

AI

START SCAN

NAVIGATION

Dashboard

Shodan Engine

Admin Panel

ADVANCED OPTIONS

Ready0%

M

Mustafa

SUPER ADMIN

Shodan Intelligence

DEV

98

QUERY CREDITS

100

SCAN CREDITS

Yes

UNLOCKED

IP Lookup

Enter IP address (e.g., 8.

Lookup

Shodan Search

port:22 country:US

Search

View search filters

DNS Resolution

example.com or IP

Resolve

Go

Exploit Search

Results

Export

Enter an IP address or search query

Results will appear here

AI Security Assistant

Select a provider below

AI PROVIDER

Ollama (Local) ✓

MODEL SELECTION

Deepseek-R1 8B

Summary

Critical

Fix Guide

Risks

Help: I'm your AI security assistant. Select a provider above to get started.

Available Providers:

Local - Private, runs on your machine

Ollama Cloud - Fast cloud inference

Gemini - Google's AI (free tier)

GitHub Models - Free GPT-4o-mini

Use Quick Actions above for common security questions!

Ask about vulnerabilities...

INTEGRATED SHODAN ENGINE

Shodan Search Filters



General

port hostname ip net org isp asn os product version

Location

country city state postal geo

HTTP

http.title http.status http.html http.component http.favicon.hash

SSL

ssl ssl.cert.subject.cn ssl.cert.issuer.cn ssl.cert.expired has_ssl

Vulnerability

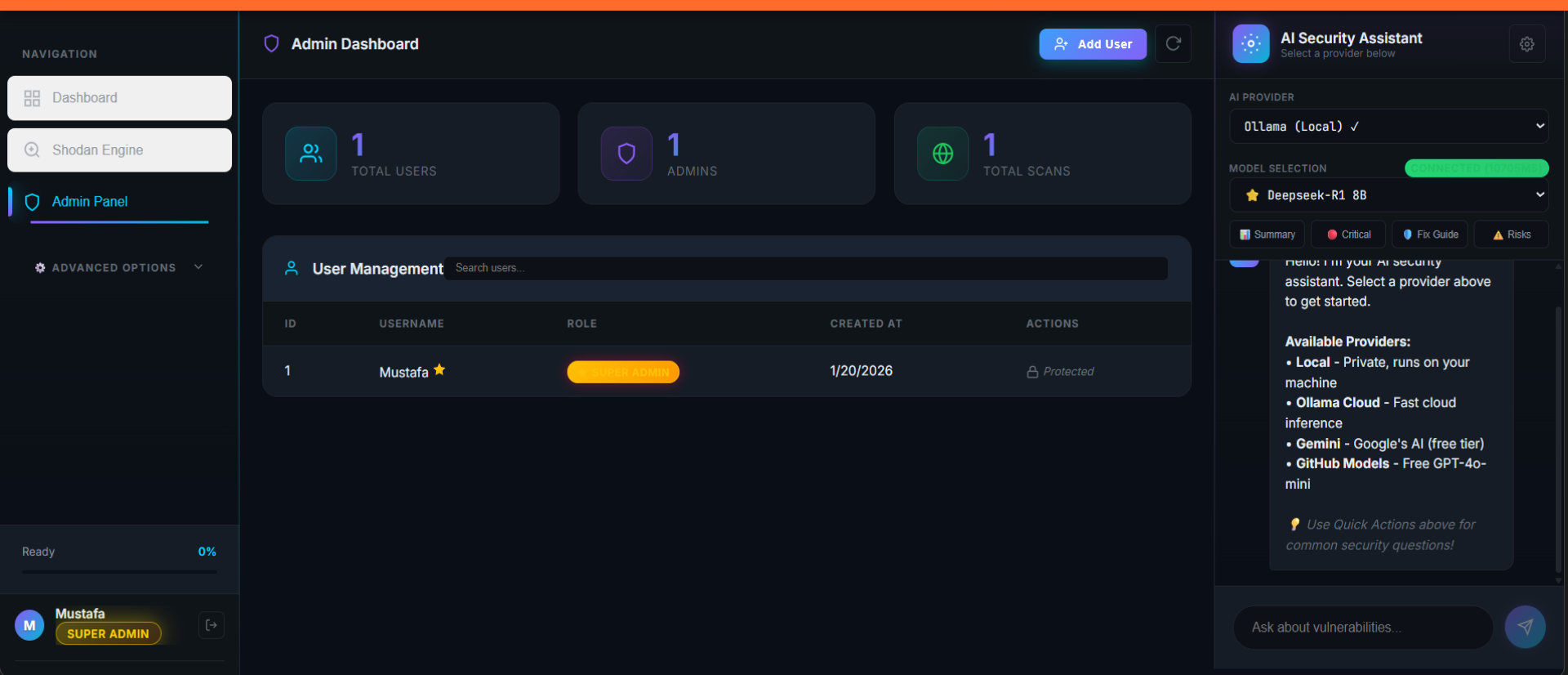
vuln cve has_vuln

Cloud

cloud.provider cloud.service cloud.region

**SHODAN ENGINE
FILTERS**

Close



ADMINISTRATION PANEL

Select previous scan...

Ready

CVE Explorer

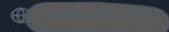
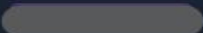
Global Vulnerability Explorer

Search CVEs...

CVE-2024-38474

CVSS 9.8

Substitution encoding issue in mod_rewrite in Apache HTTP Server 2.4.59 and earlier allows attacker to execute scripts in...

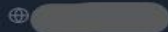
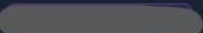


1/20/2026

CVE-2024-38474

CVSS 9.8

Substitution encoding issue in mod_rewrite in Apache HTTP Server 2.4.59 and earlier allows attacker to execute scripts in...

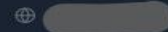
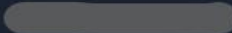


1/20/2026

CVE-2024-38474

CVSS 9.8

Substitution encoding issue in mod_rewrite in Apache HTTP Server 2.4.59 and earlier allows attacker to execute scripts in...

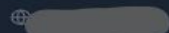
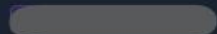


1/20/2026

CVE-2024-38474

CVSS 9.8

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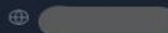
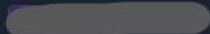


1/20/2026

CVE-2024-38474

CVSS 9.8

Substitution encoding issue in mod_rewrite in Apache HTTP Server 2.4.59 and earlier allows attacker to execute scripts in...

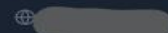
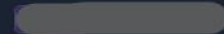


1/20/2026

CVE-2024-38474

CVSS 9.8

Substitution encoding issue in mod_rewrite in Apache HTTP Server 2.4.59 and earlier allows attacker to execute scripts in...



1/20/2026

CVE-2024-38474

CVSS 9.8

CVE-2024-38474

CVSS 9.8

CVE-2024-38476

CVSS 9.8

CVE EXPLORERE

TwinSanity Recon

> target.com

Subs

Shodan

AI

START SCAN

AI Security Assistant

AI Analysis

NAVIGATION

Dashboard

Shodan Engine

Admin Panel

ADVANCED OPTIONS

Select previous scan...

Ready

34

IP ADDRESSES

75

SUBDOMAINS

97

UNIQUE CVEs

17

CRITICAL

Findings Report

AI Analysis Report

Tools Findings

2782

VULNERABILITY SEVERITY

Critical

High

Medium

Low

TOP OPEN PORTS

88

443

8080

2082

2083

8443

2086

2087

0

5

10

15

20

25

30

35

Scan complete!

100%

M

Mustafa

SUPER ADMIN

Discovered Subdomains

75

Filter subdomains...

AI PROVIDER

Ollama (Local) ✓

MODEL SELECTION

Deepseek-R1 8B

Summary

Critical

Fix Guide

Risks

Scan results for [redacted] infrastructure reveal several high-risk Apache HTTP Server instances running outdated versions with multiple **CRITICAL** CVEs (CVSS 9.8) affecting multiple IPs. The most **CRITICAL** findings include servers vulnerable to remote code execution, SSRF attacks, and information disclosure vulnerabilities. One server also shows potential HTTP/2 denial of service vulnerability.

Ask about vulnerabilities...

SCAN RESULTS VIEW



Executive Summary

Scan results reveal multiple critical Apache HTTP Server vulnerabilities affecting several [REDACTED] subdomains, along with exposed MySQL databases and end-of-life systems with known exploits. The most severe findings include CVSS 9.8 vulnerabilities in Apache mod_rewrite allowing remote code execution.

Scan results for [REDACTED] infrastructure reveal several high-risk Apache HTTP Server instances running outdated versions with multiple critical CVEs (CVSS 9.8) affecting multiple IPs. The most critical findings include servers vulnerable to remote code execution, SSRF attacks, and information disclosure vulnerabilities. One server also shows potential HTTP/2 denial of service vulnerability.



High-Risk Assets

AI EXECUTIVE SUMMARY OF FINDINGS



HOST INFORMATION

IP ADDRESS

OPEN PORTS

80, 443

TECHNOLOGIES

None detected

VULNERABILITY SUMMARY

4

CRITICAL

20

HIGH

11

MEDIUM

2

LOW

CVE Details (37) Filter CVEs...

CVE-2024-38474

9.8 CRITICAL

Substitution encoding issue in mod_rewrite in Apache HTTP Server 2.4.59 and earlier allows attacker to execute scripts in directories permitted by the configuration but not directly reachable by any URL or source disclosure of scripts meant to only to be executed as CGI. Users are recommended to upgrade to version 2.4.60, which fixes this issue. Some RewriteRules that...

CVE-2024-38476

9.8 CRITICAL

Vulnerability in core of Apache HTTP Server 2.4.59 and earlier are vulnerably to information disclosure, SSRF or local script execution via backend applications whose response headers are malicious or exploitable. Users are recommended to upgrade to version 2.4.60, which fixes this issue.

CVE-2024-38475

9.1 CRITICAL

INFORMATION
ABOUT SUBDOMAIN

CVE Details (37) 2023

CVE-2024-42516

7.5 HIGH

HTTP response splitting in the core of Apache HTTP Server allows an attacker who can manipulate the Content-Type response headers of applications hosted or proxied by the server can split the HTTP response. This vulnerability was described as CVE-2023-38709 but the patch included in Apache HTTP Server 2.4.59 did not address the issue. Users are recommended to upgrade to...

CVE-2023-38709

7.3 HIGH

Faulty input validation in the core of Apache allows malicious or exploitable backend/content generators to split HTTP responses. This issue affects Apache HTTP Server: through 2.4.58.

SEARCH FOR SPECIFIC CVE

CVE Details (37) 2023

CVE-2024-42516

7.5 HIGH

HTTP response splitting in the core of Apache HTTP Server allows an attacker who can manipulate the Content-Type response headers of applications hosted or proxied by the server can split the HTTP response. This vulnerability was described as CVE-2023-38709 but the patch included in Apache HTTP Server 2.4.59 did not address the issue. Users are recommended to upgrade to...

NIST

Information Technology Laboratory

NATIONAL VULNERABILITY DATABASE

VULNERABILITIES

CVE-2024-42516 Detail

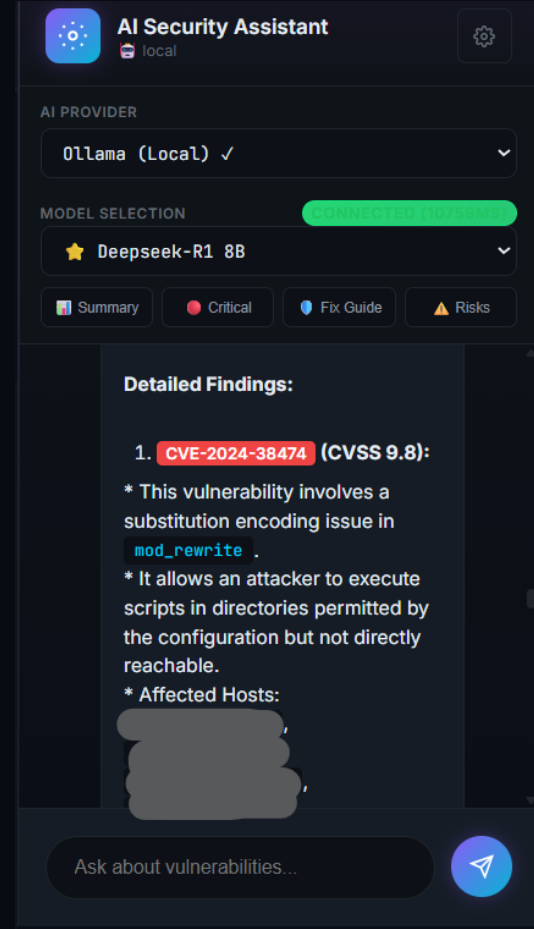
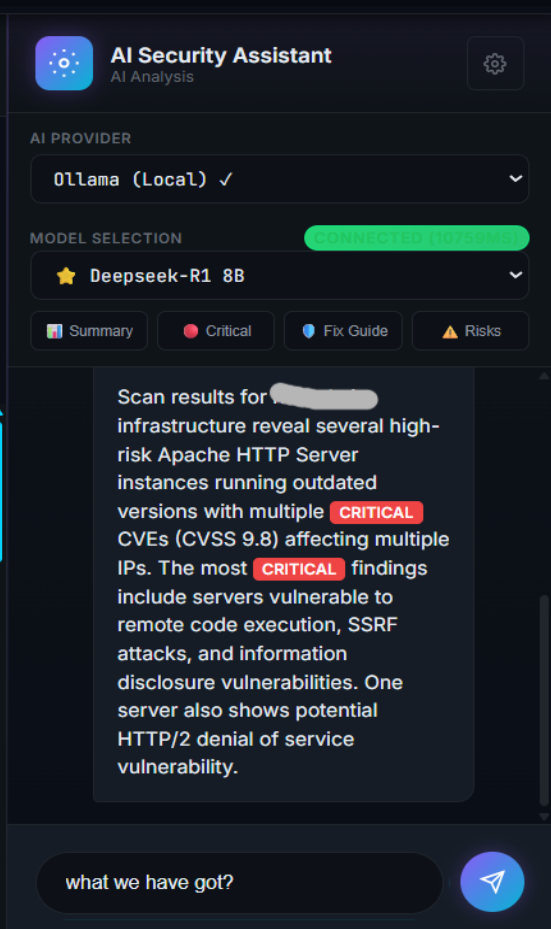
MODIFIED

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

HTTP response splitting in the core of Apache HTTP Server allows an attacker who can manipulate the Content-Type response headers of applications hosted or proxied by the server can split the HTTP response. This vulnerability was described as CVE-2023-38709 but the patch

VULNERABILITY NIST REFERENCE



AI CHAT ASSISTS

1

Dashboard header and summary cards.

Header: [Lock icon] [Redacted] [Ready] [Icons: Home, Global, Add, Refresh, Delete, Undo, Redo, Print]

Summary Cards:

- 34 IP ADDRESSES
- 75 SUBDOMAINS
- 97 UNIQUE CVES
- 17 CRITICAL

Right Sidebar:

- AI PROVIDER: Ollama (Local)
- MODEL SELECTION: ★ Deepseek-R1

2

 **Generate Report**

×

REPORT FORMAT

☒ <> HTML

☐  PDF

☒ Include AI Security Analysis

☒ Include Tool Findings (Nuclei, XSS, API)

Cancel

 **Generate & Download**

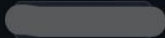
REPORT GENERATION



TwinSanity Security Report

Automated Vulnerability Assessment

TARGET DOMAIN



SCAN ID

4f7ab79d

GENERATED

2026-01-21 23:36:54

IPS ANALYZED

34



CRITICAL



34

IPS SCANNED



75

SUBDOMAINS



97

UNIQUE CVEs



17

CRITICAL



45

HIGH



32

MEDIUM



3

LOW

HTML REPORT VIEW



Executive Summary

Scan results reveal multiple critical Apache HTTP Server vulnerabilities affecting several [REDACTED] subdomains, along with exposed MySQL databases and end-of-life systems with known exploits. The most severe findings include CVSS 9.8 vulnerabilities in Apache mod_rewrite allowing remote code execution. Scan results for [REDACTED] infrastructure reveal several high-risk Apache HTTP Server instances running outdated versions with multiple critical CVEs (CVSS 9.8) affecting multiple IPs. The most critical findings include servers vulnerable to remote code execution, SSRF attacks, and information disclosure vulnerabilities. One server also shows potential HTTP/2 denial of service vulnerability.



High Risk Assets

IP ADDRESS	HOSTNAME	SEVERITY	RISK REASON
[REDACTED]	[REDACTED]	CRITICAL	Multiple high-severity Apache vulnerabilities including CVSS 9.8 RCE in mod_rewrite and exposed web services
[REDACTED]	[REDACTED]	CRITICAL	Multiple critical Apache vulnerabilities (CVSS 9.8) and extensive web service exposure
[REDACTED]	[REDACTED]	CRITICAL	Critical Apache vulnerabilities (CVSS 9.8) and cloud-hosted infrastructure



SELECT MISSION

[REDACTED] - 1/20/2026 [COMPLETED] ▼

Target: [REDACTED] Scanned: 1/20/2026, 4:41:19 PM Status: Completed



34

ISLANDS



201

OPEN PORTS



429

CVEs FOUND



64

CRITICAL



22

CLEAN HOSTS

DISCOVERED ISLANDS



1 CVE



SAFE



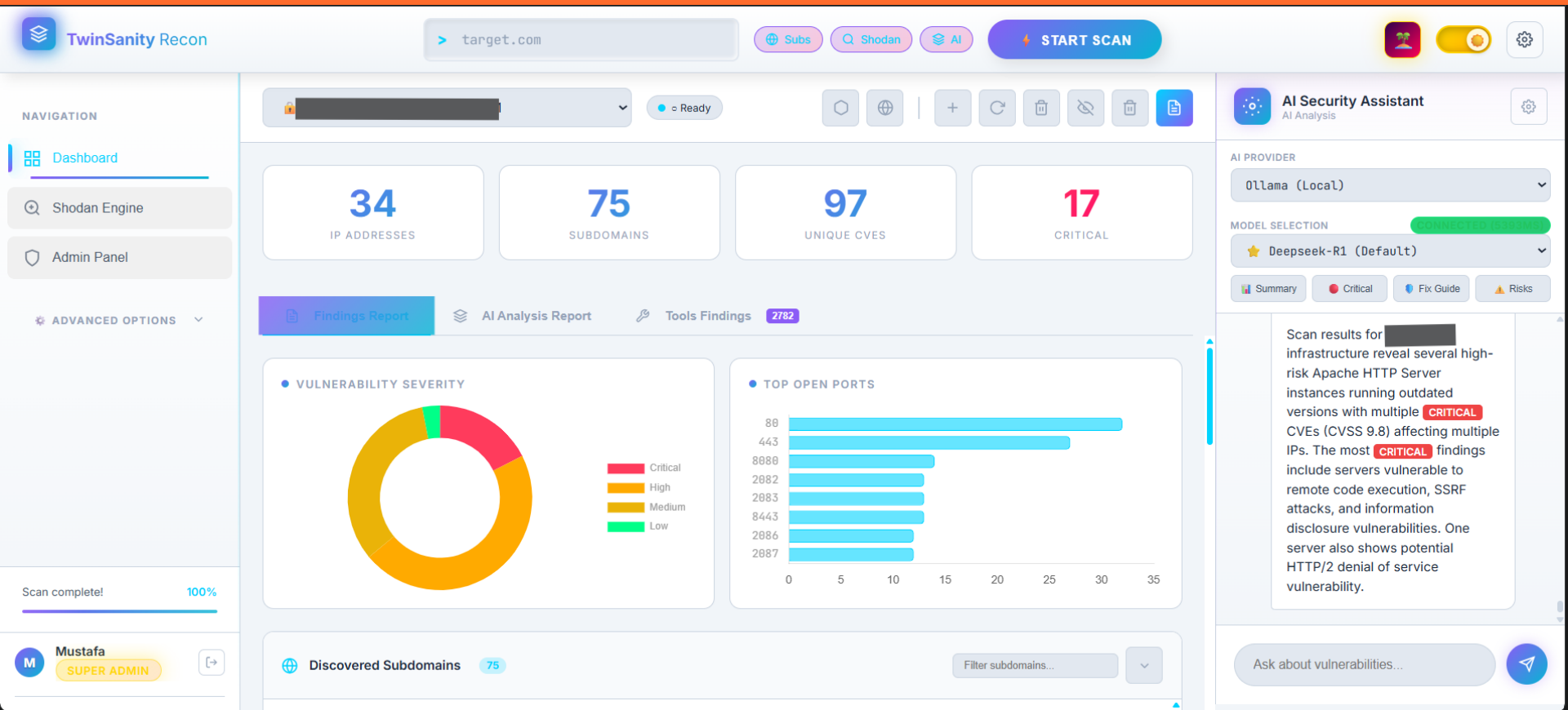
SAFE



37 CVEs

CRITICAL

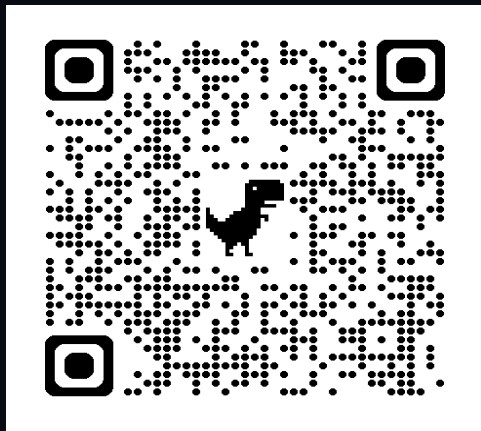
TWINSANITY ISLAND VIEW



LIGHT SUPPORTED

Ready to Hunt?

Start your reconnaissance journey with TwinSanity Recon



Notion Documentation

GitHub Repository

Telegram Channel

